AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 4 · WORKBOOK

Python and Table Readiness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Keep one readable contract-and-summary record.

Trace schema checks, units, missingness and bounded summary conclusions.

Keep one habit: Core W1–W3/W5 → selected independent task → project evidence; optional file/parser tasks have separate keys.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

Classification or Regression?

> **My remembered idea, example and correction**

The ML Lifecycle

> **My remembered idea, example and correction**

MAIN LESSON · SECTION 12

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Core schema: activity nonempty text, minutes exact int 0..60 inclusive; required keys assumed by the original script. Reading/12 and Sorting/0 print True. W2 minutes 0,60 pass; -1,61 fail. W3 source 180 seconds converts to 3 minutes with source/conversion retained. W5 valid Reading/12 does not establish truth.

  Separate pictured heights P=2,Q=6,R=None,S=0 cm give sum 8, mean 8/3 cm and coverage 3/4; S becoming None gives mean 4 cm and 2/4 coverage.

- Reference for this exercise: open this week's student lesson, "Python and Table Readiness", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- List the required fields, **type**s and duration boundaries for the worked table.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Core schema: activity nonempty text, minutes exact int 0..60 inclusive; required keys assumed by the original script. Reading/12 and Sorting/0 print True. W2 minutes 0,60 pass; -1,61 fail. W3 source 180 seconds converts to 3 minutes with source/conversion retained. W5 valid Reading/12 does not establish truth.

  Separate pictured heights P=2,Q=6,R=None,S=0 cm give sum 8, mean 8/3 cm and coverage 3/4; S becoming None gives mean 4 cm and 2/4 coverage.

- Classify minutes values 0, 60, -1 and 61 under this **schema**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Core schema: activity nonempty text, minutes exact int 0..60 inclusive; required keys assumed by the original script. Reading/12 and Sorting/0 print True. W2 minutes 0,60 pass; -1,61 fail. W3 source 180 seconds converts to 3 minutes with source/conversion retained. W5 valid Reading/12 does not establish truth.

  Separate pictured heights P=2,Q=6,R=None,S=0 cm give sum 8, mean 8/3 cm and coverage 3/4; S becoming None gives mean 4 cm and 2/4 coverage.

- A source says 180 seconds but the row says minutes: 180. Give a documented correction.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Core schema: activity nonempty text, minutes exact int 0..60 inclusive; required keys assumed by the original script. Reading/12 and Sorting/0 print True. W2 minutes 0,60 pass; -1,61 fail. W3 source 180 seconds converts to 3 minutes with source/conversion retained. W5 valid Reading/12 does not establish truth.

  Separate pictured heights P=2,Q=6,R=None,S=0 cm give sum 8, mean 8/3 cm and coverage 3/4; S becoming None gives mean 4 cm and 2/4 coverage.

- A row says Reading, 12 and passes. Does that establish the session really lasted **12 minutes**?

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the three supplied values as the observed records. Treat the missing fourth record as unknown, not zero. State the policy and keep observed-count and total-record-count separate. For an empty collection, explain the status before attempting division; no new measurement is supplied.

- Supplied values are3,7,14. Compute their mean. What should a function report for an empty list? How should it handle a missing fourth record?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Changed independent case

My independent explanation, trace and limits

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a tiny five-row plant table with explicit column names, **type**s, **unit**s, and a reproducible summary calculation.
- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 27

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.

- I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Classify records for Rows columns and **schema**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for Rows columns and **schema**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Trace mean for [4,8,12] using sum(values)/len(values). What should the program do for []? Explain why returning 0 silently is misleading.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Python lengths=[4,6,8]; total=sum(lengths); mean=total/len(lengths).
- Trace and handle an empty list.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Trace Python: lengths = [4, 7, 9]; selected = lengths[1]; total = selected + 3. Give selected and total.
- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 4 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Trace mean for [4,8,12] using sum(values)/len(values). What should the program do for []? Explain why returning 0 silently is misleading.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

My reasoning and evidence

▼ **Extension · File input and output — W1 Trace types**

REFERENCE · SECTION 17

Extension · File input and output — W1 Trace types

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Before and after int("8"), identify the **type**s and values.

> **My reasoning and evidence**

▼ **Extension · File input and output — W2 Predict the total**

REFERENCE · SECTION 18

Extension · File input and output — W2 Predict the total

> Retained extension task with its own values and teacher key.

Extension question and evidence

- A sandbox **file** has lines 5, 0 and 2. What total does the worked reader print?

My reasoning and evidence

▼ **Extension · File input and output — W3 Repair a path issue**

REFERENCE · SECTION 19

Extension · File input and output — W3 Repair a path issue

> Retained extension task with its own values and teacher key.

Extension question and evidence

- The **file** is in the class-data folder but the script looks in its working folder. Give a safe repair.

My reasoning and evidence

▼ **Extension · File input and output — W4 Read independently**

REFERENCE · SECTION 20

Extension · File input and output — W4 Read independently

> Retained extension task with its own values and teacher key.

Extension question and evidence

- In a fresh sandbox **file** counters.txt put 4, 6 and 0 on separate lines. Adapt the reader, print the number of values and total. State the expected results.

> **My reasoning and evidence**

▼ **Extension · File input and output — W5 Handle malformed data**

REFERENCE · SECTION 21

Extension · File input and output — W5 Handle malformed data

> Retained extension task with its own values and teacher key.

Extension question and evidence

- The **file** contains twelve on one line. Why should the program not silently turn it into zero?

My reasoning and evidence

▼ Extension · Exceptions and recovery — W1 Trace success

REFERENCE · SECTION 22

Extension · Exceptions and recovery — W1 Trace success

> Retained extension task with its own values and teacher key.

Extension question and evidence

- What does parse_minutes("0") return, and why is zero allowed?

My reasoning and evidence

▼ Extension · Exceptions and recovery — W2 Trace two failures

REFERENCE · SECTION 23

Extension · Exceptions and recovery — W2 Trace two failures

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Compare parse_minutes("abc") and parse_minutes("70").

My reasoning and evidence

▼ **Extension · Exceptions and recovery — W3 Repair a handler**

REFERENCE · SECTION 24

Extension · Exceptions and recovery — W3 Repair a handler

> Retained extension task with its own values and teacher key.

Extension question and evidence

- An except block returns 0 for any failure. State two problems and a repair.

My reasoning and evidence

▼ **Extension · Exceptions and recovery — W4 Build independently**

REFERENCE · SECTION 25

Extension · Exceptions and recovery — W4 Build independently

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Adapt the function for count strings representing integers 0 through 10. Test "4", "0", "11", "four" and "-1" and record **status**/value.

My reasoning and evidence

▼ **Extension · Exceptions and recovery — W5 Choose recovery**

REFERENCE · SECTION 26

Extension · Exceptions and recovery — W5 Choose recovery

> Retained extension task with its own values and teacher key.

Extension question and evidence

- A batch has 8 valid and 2 invalid lines. What should its report include?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 28

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Inspect a small table before asking Python to compute an average.

- Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/4

Python and Table Readiness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Check records

Four rows contain three measured values and one missing value.

2. Calculate mean

Measured sum is 2+6+0=8; mean is 8/3 cm, approximately 2.67 cm.

3. Report coverage

Report 3/4 measured, retaining the missing record separately.

CHANGE ONE THING

S changes from 0 to None.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 29

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 30

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Inspect a small table before asking Python to compute an average.
- Which labelled step matches this detail? Report 3/4 measured, retaining the missing record separately.
- 1. Check records
- 2. Report coverage
- 3. Calculate mean
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Report coverage Report 3/4 measured, retaining the missing record separately. Read the picture in order. Report 3/4 measured, retaining the missing record separately.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: S changes from 0 to None. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Mean becomes (2+6)/2=4 cm with 2/4 measured.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Mean becomes (2+6)/2=4 cm with 2/4 measured. Removing a measured zero changes both the denominator and the coverage.

Repair and keep the record

Review Classification or Regression? (ai-4/3). Goal: Identify a task from the meaning of its target. Short practice: Did regression MAE improve when the boundary changed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review The ML Lifecycle (ai-4/1). Goal: Distinguish **fitting** a rule from applying a fixed rule. Short practice: Should the team silently retune on the changed test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

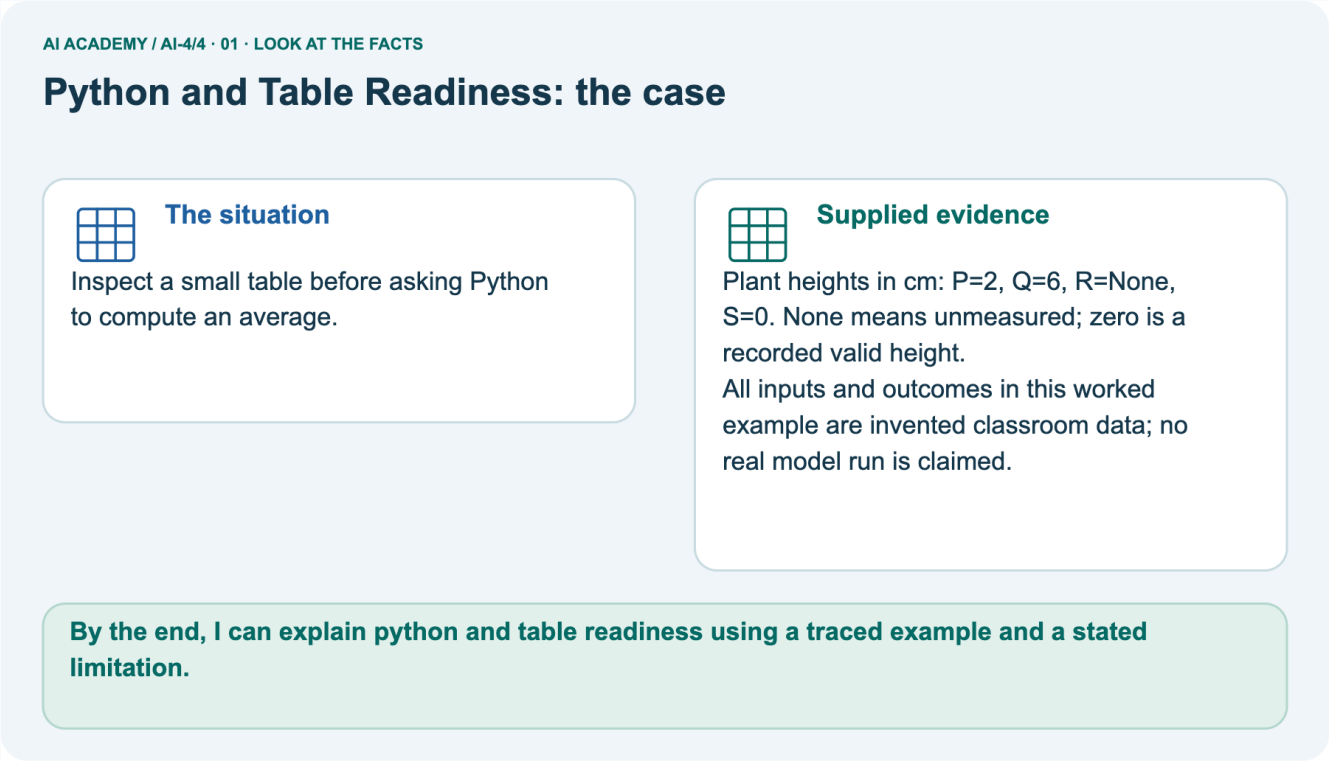
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 31

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Python and Table Readiness: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

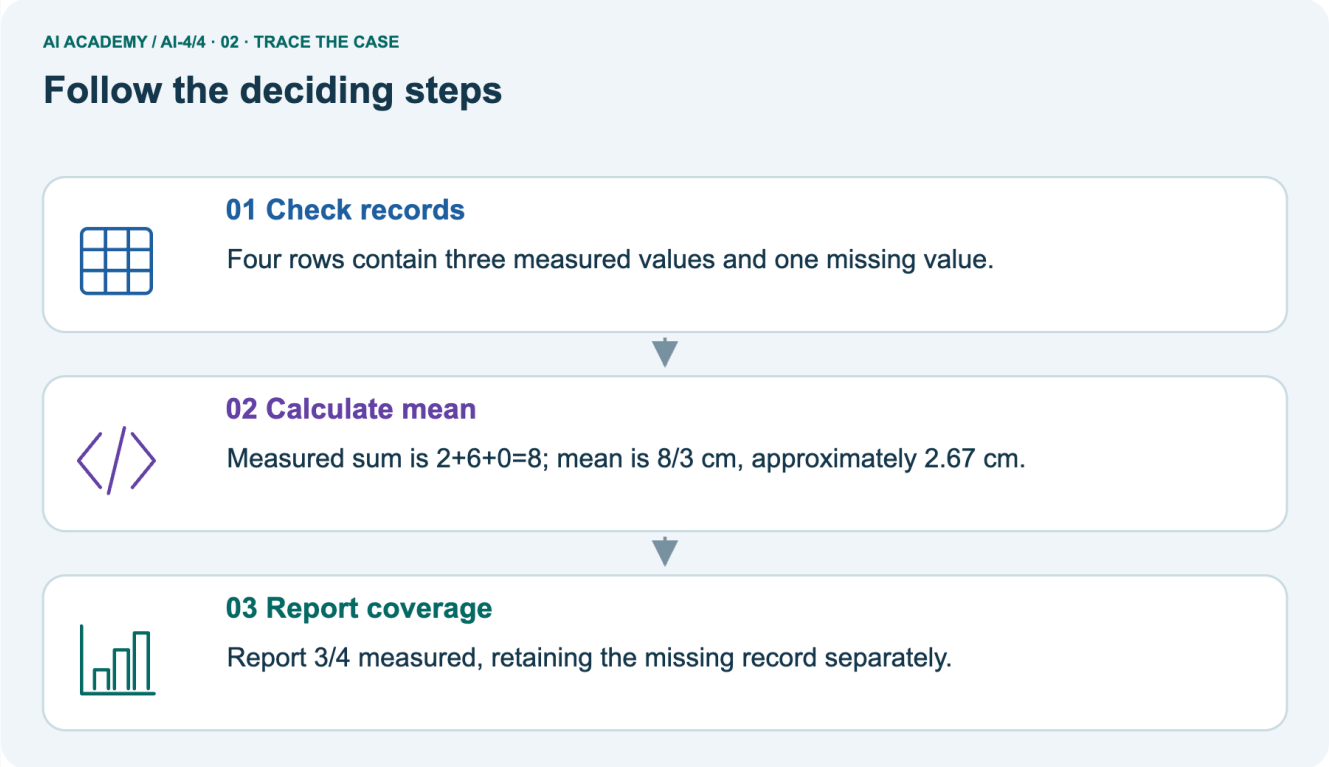
Inspect a small table before asking Python to compute an average. Supplied facts: Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



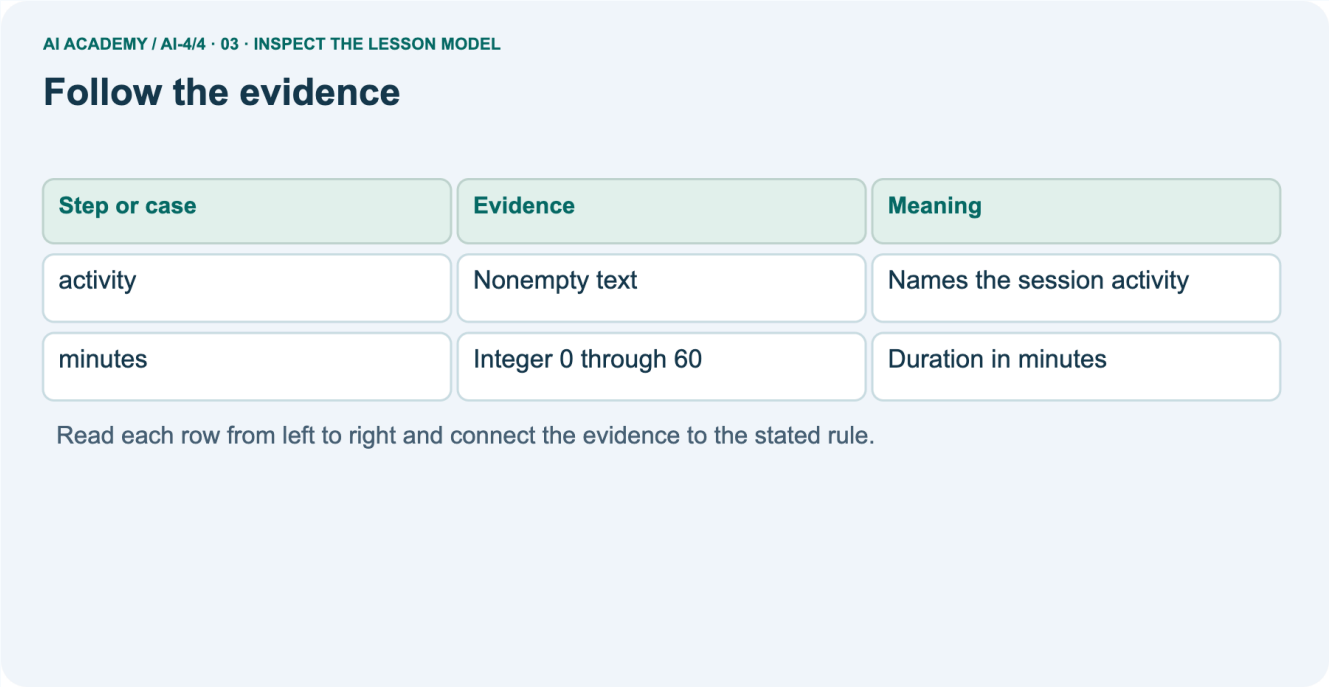
Follow the labelled picture with the matching explanation. Open the vector diagram

Check records: Four rows contain three measured values and one missing value. → Calculate mean: Measured sum is 2+6+0=8; mean is 8/3 cm, approximately 2.67 cm. → Report coverage: Report 3/4 measured, retaining the missing record separately.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence



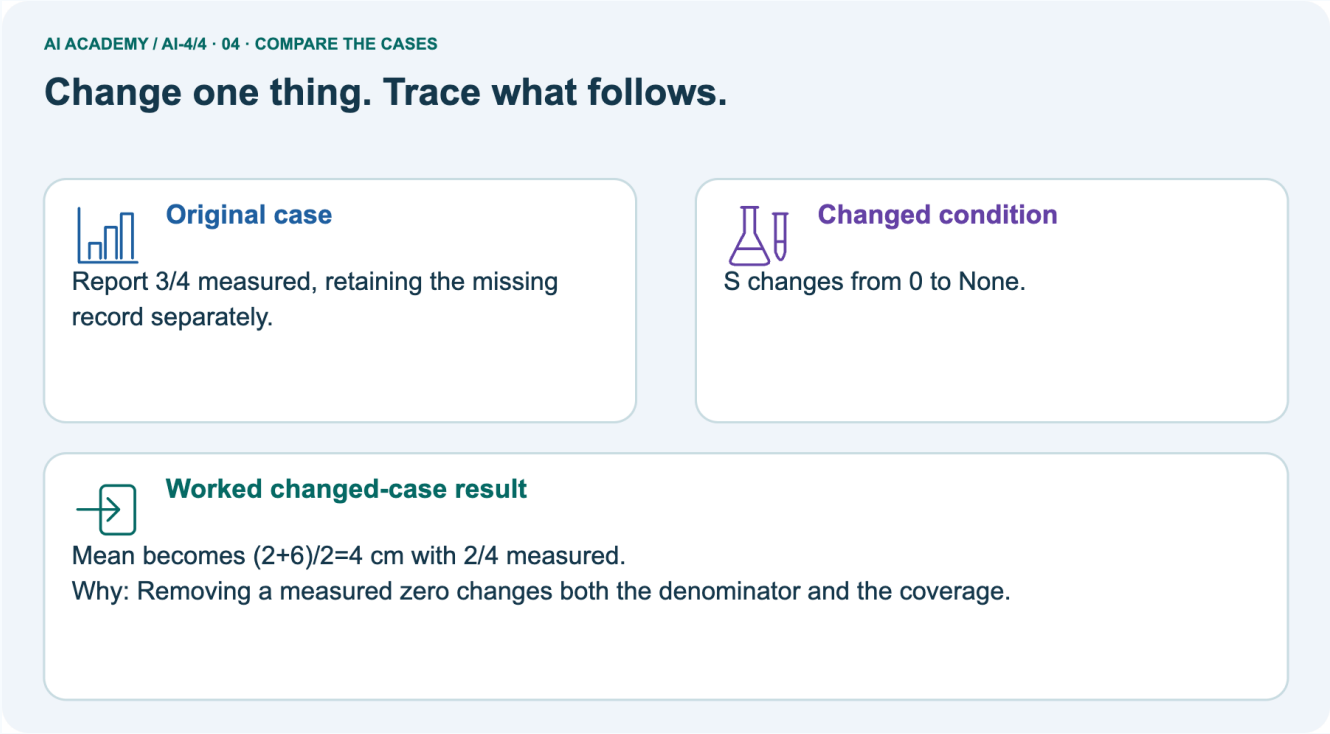
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. activity | Nonempty text | Names the session activity. minutes | Integer 0 through 60 | Duration in minutes

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

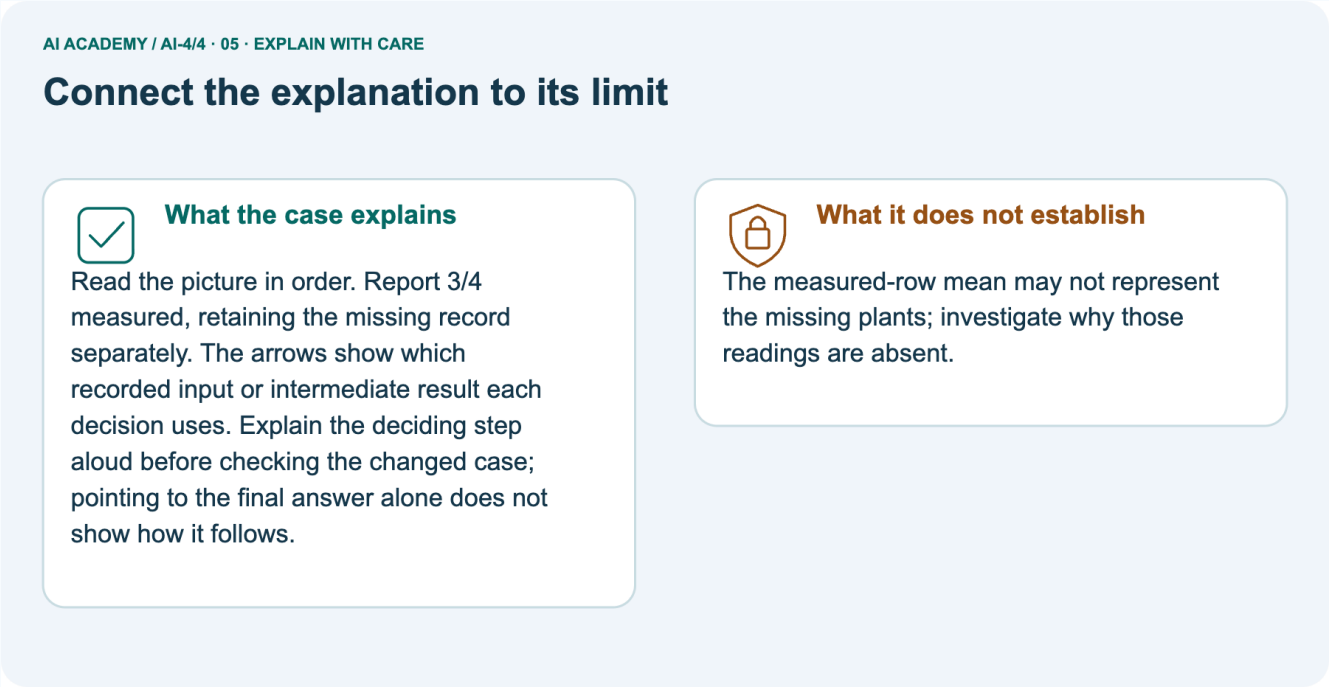
Original case: Report 3/4 measured, retaining the missing record separately. Changed condition: S changes from 0 to None. Expected result: Mean becomes (2+6)/2=4 cm with 2/4 measured.

Reason: Removing a measured zero changes both the denominator and the coverage.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Report 3/4 measured, retaining the missing record separately. The arrows show which recorded **input** or intermediate result each decision uses.

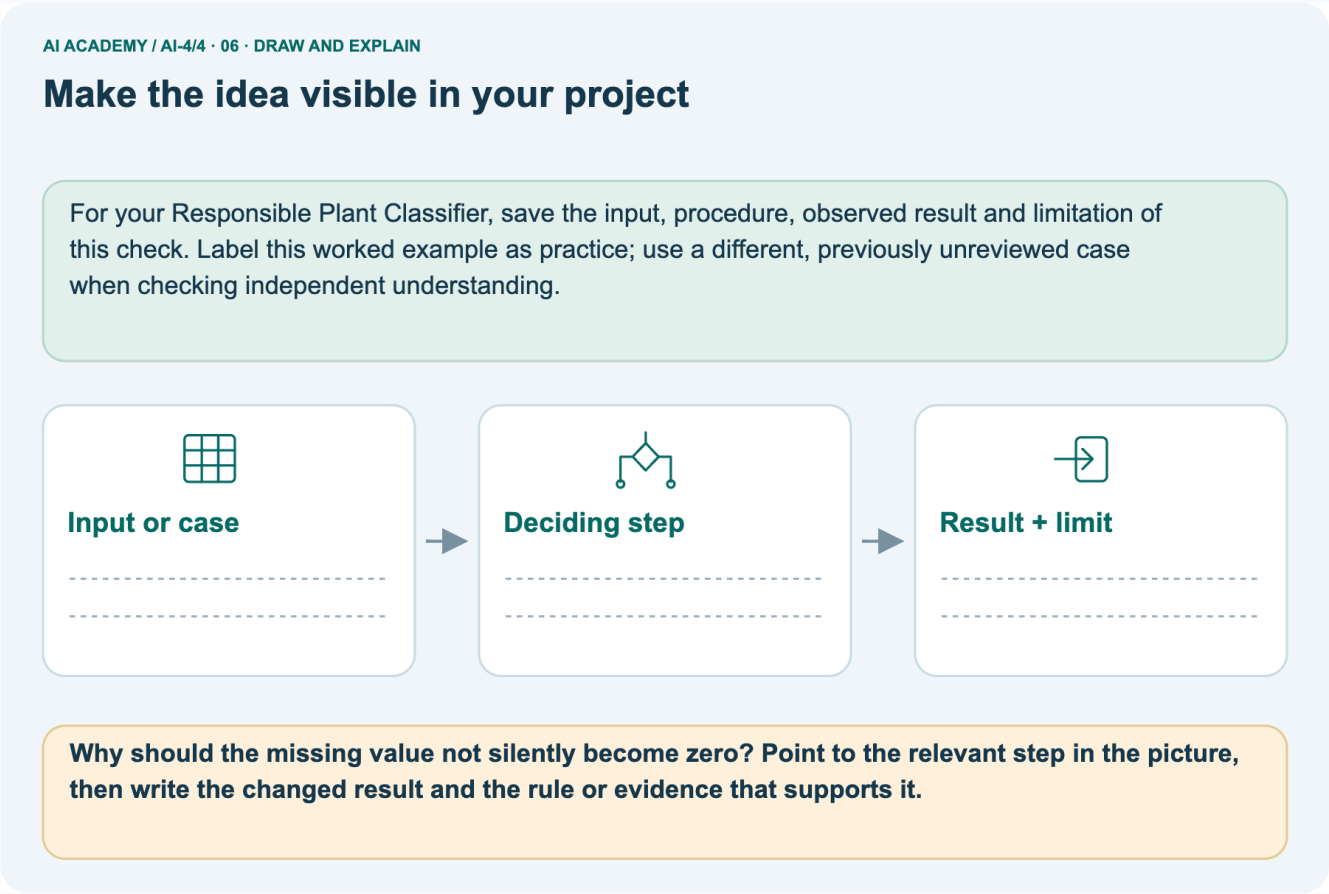
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: The measured-row mean may not represent the missing plants; investigate why those readings are absent.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Python and Table Readiness: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
</details>